

From Biological Axons to Artificial Neurons: A Computational Comparison

Analyzing Biophysical Signaling vs Mathematical Activation Functions

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Independent Scientific Review Notes | Date: September 12, 2023

Abstract

Artificial neural networks were originally conceptualized as mathematical abstractions of biological nervous systems. This review compares biological neuronal signaling, action potentials, and synaptic transmission with artificial perceptron models, weight optimization, and error backpropagation.

1. Historical Foundations of Artificial Neurons

The conceptual origin of artificial intelligence is deeply rooted in neurophysiology. In 1943, Warren McCulloch and Walter Pitts proposed a simplified binary model of a biological neuron. Frank Rosenblatt expanded this in 1958 with the Perceptron, establishing the precursor to modern artificial neural network architectures.

2. Comparative Biophysics: Action Potentials vs Matrix Operations

Biological signal transmission relies on ionic gradients across neuronal membranes. Upon reaching depolarization threshold, voltage-gated sodium channels open, generating self-propagating action potentials along myelinated axons. Synaptic transmission converts electrical signals into chemical neurotransmitters across a 20-nanometer synaptic cleft.

By contrast, artificial neurons reduce biological complexity to linear algebra: computing the inner product of input vectors and weight vectors, adding a scalar bias, and applying a non-linear activation function (e.g., Rectified Linear Unit [ReLU] or Sigmoid).

3. Learning Paradigms: Activity Plasticity vs Backpropagation

Biological networks modify synaptic connections locally through activity-dependent mechanisms such as Long-Term Potentiation (LTP). In contrast, artificial networks update connections globally via error backpropagation, using partial derivatives calculated via the chain rule to minimize cost functions through stochastic gradient descent.

4. Energetic Efficacy and Neuromorphic Computing

The human brain operates on approximately 20 watts of power while executing complex sensory, motor, and cognitive tasks. Deep artificial networks require kilowatts to megawatts for training. Emerging neuromorphic architectures, including event-driven spiking neural networks, seek to bridge this computational efficiency gap.

References

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